10.5 Infinitely Many Bounces in Finite Time

WORKSHEET

The question

A particle moves on a smooth triangular horizontal surface AOB with angle $AOB = 30°$. The surface is bounded by two vertical walls OA and OB and the coefficient of restitution between the particle and the walls is e, where $e < 1$. The particle, which is initially at point P on the surface and moving with velocity u_1, strikes the wall OA at M_1, with angle $PM_1A = \theta$, and rebounds, with velocity v_1, to strike the wall OB at N_1, with angle $M_1N_1B = \theta$. Find e and $\dfrac{v_1}{u_1}$ in terms of θ.

The motion continues, with the particle striking side OA at $M_2, M_3, \ldots$ and striking side OB at $N_2, N_3, \ldots$. Show that, if $\theta < 60°$, the particle reaches O in a finite time.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself When a particle hits a smooth wall with coefficient of restitution e, what happens to the component of velocity along the wall? And the component perpendicular to it?

STAGE 1 The angle of departure

Ask yourself The particle arrives at M_1 at angle θ to the wall OA (measured on the side away from O). At what angle φ to OA does it leave? Use the triangle OM_1N_1 to find φ in terms of θ.

STAGE 2 The pattern repeats

Ask yourself At N_1, the particle arrives at angle θ to the wall OB, measured on the side away from O. Compare this with its arrival at M_1. What happens next?

STAGE 3 Two ratios

Ask yourself By what factor ρ does the distance from O shrink from one bounce to the next? By what factor k does the speed shrink? So by what factor does the time for each leg shrink?

STAGE 4 A geometric series

Ask yourself When is $\sin(2\theta - 60^\circ) < \sin 2\theta$? Use the identity $\sin X - \sin Y = 2\cos\frac{1}{2}(X+Y)\sin\frac{1}{2}(X-Y)$.

Try this next

1 **The total distance.** Show that the total distance travelled from M_1 to O is finite for every θ between 30° and 90°, and find it in terms of OM_1 and θ.

2 **Other wedges.** Repeat the question for a wedge of angle α instead of 30°. Show that $\varphi = \theta - \alpha$, and find the condition on θ for the particle to reach O in a finite time.

3 **Elastic walls.** If $e = 1$, can the particle ever reach O? (*Hint:* reflect the wedge repeatedly in its walls, so that the path becomes a straight line. How many copies

of a 30° wedge fit round O?)

4 The bouncing ball. A ball is dropped from height 1 m with $e = 0.8$. How long does it bounce for, and how far does it travel in total? (Take $g = 9.8\ \mathrm{m\,s^{-2}}$.)

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 5. Includes several problems on bouncing balls and infinite sequences of collisions.
- **Sean McNamara & William Young, "Inelastic collapse and clumping in a one-dimensional granular medium"**, *Physics of Fluids A* 4 (1992).
- **Serge Tabachnikov, *Geometry and Billiards*** (American Mathematical Society; free online). The mathematics of billiards, including the unfolding trick in the exercises.

The full guided solution, and what the question leads to at university, is in Section 10.5:
https://beyondstep.keithmcnulty.org/ch10/zeno-wedge.html